

THE NEW SCHOOL • EUGENE LANG COLLEGE OF LIBERAL ARTS

Calculus

A Multidisciplinary Approach

LMTH 2040 • Section TBD • CRN TBD

Fall 2026 • Mondays & Wednesdays • 12:00–1:40 PM • TBD

Course Description

Calculus is a way of reasoning about accumulation, change, dynamics, and form. This course follows the historical and conceptual development of these ideas from problems of measurement, quadrature, and motion to differential equations, high-dimensional optimization, and contemporary artificial intelligence. Mathematics is studied alongside computation, visual culture, modeling, history and sociology of science, and critical studies of data and AI.

Rather than treating history or culture as decoration around a fixed mathematical sequence, the course takes what Otto Toeplitz called a *genetic approach*: problems and practices that motivated mathematical ideas help introduce those ideas. Students will move repeatedly among concrete problems, finite constructions, computational experiments, visualizations, formal mathematics, models, and critical interpretation.

The course begins with integration rather than differentiation. This order reflects both the historical development of calculus and the course's emphasis on measurement and accumulation. Problems of area, volume, summation, and infinity provide a route into the integral before we turn to motion, tangency, and the derivative. These two traditions eventually meet in the Fundamental Theorem of Calculus.

Guiding Questions

- How can finite procedures be used to describe continuous quantities?
- How do approximation and limiting processes produce mathematical knowledge?
- How do we represent and measure change?
- What can mathematical models reveal, and what do they conceal?
- How did calculus become a technology of prediction, optimization, and control?
- What happens when people, images, language, and intelligence become mathematical representations?

Course Architecture

The semester is organized into four units, each containing two modules.

Unit	Modules	Central Movement
I. Integration	1. Measurement, Exhaustion & Infinity; 2. Integration, Probability & Data	accumulation, approximation, quantification
II. Differentiation	3. Motion, Tangency & the Derivative; 4. Optimization, Networks & Learning	change, composition, optimization
III. Differential Equations	5. Dynamics, Prediction & Simulation; 6. Feedback, Control & Intelligence	rules of change, simulation, feedback
IV. Higher Dimensions & PDEs	7. Surfaces, Fields & Partial Differential Equations; 8. Images, Representation & Generative AI	change across dimensions, space, and time

Calculus Colloquium

Throughout the semester students will open selected classes with short scholarly presentations extending the current material. Each colloquium contribution consists of a **three-page LaTeX brief**, **five slides**, and a **5–7 minute presentation** followed by discussion.

Topics may include an additional reading, historical episode, mathematical problem, visual object, model, computational experiment, or contemporary application. Across successive rounds the emphasis will move from **explanation** to **analysis** to **synthesis**. Selected work will be revised after feedback.

The Colloquium is intended to develop mathematical communication as well as mathematical knowledge: students will practice explaining technical ideas precisely, constructing mathematical arguments, designing effective visual representations, and presenting technical work to an interdisciplinary audience.

Projects

Project I: Modeling, Measurement & Representation. A midsemester investigation combining calculus, data or observation, visualization, computation, and interpretation. Students will formulate a question, determine what can be measured or represented mathematically, construct an analysis, and examine the assumptions involved in that representation.

Final Project: Calculus in the World. A sustained mathematical and computational investigation connecting calculus to a historical, cultural, scientific, social, technological, or artistic question. Projects will culminate in a written technical report and presentation at the final symposium.

Readings and Seminar

Calculus has a history. Its concepts emerged from particular problems, arguments, instruments, institutions, and disagreements about what constitutes mathematical knowledge. We will therefore read mathematics alongside texts from the history and sociology of science, science and technology studies, visual culture, cybernetics, data studies, and critical studies of artificial intelligence.

Historical primary sources will include selections from Archimedes, Bonaventura Cavalieri, Isaac Barrow, Gottfried Wilhelm Leibniz, Jean le Rond d'Alembert, Leonhard Euler, Daniel Bernoulli, and Gaspard Monge.

Many of these readings will be drawn from Dirk Struik's *A Source Book in Mathematics, 1200–1800*.

Other readings will include selections from Otto Toeplitz, Michel Serres, Gilles Deleuze and Félix Guattari, Theodore Porter, Lorraine Daston and Peter Galison, Norbert Wiener, Andrew Pickering, Geoffrey Bowker and Susan Leigh Star, Trevor Paglen, and Kate Crawford, together with primary sources from the history of neural networks and machine learning.

These texts are not assigned as authorities to be accepted. Students will be asked to reconstruct arguments, identify assumptions, compare different forms of evidence, and evaluate claims in relation to the mathematics we are studying.

Calendar

Date	Module	Topic	Colloquium Project /
Aug 26	Opening	What is calculus? Measurement, approximation, infinity, computation, and the genetic approach	Course introduction
Aug 31	1	Patterns and accumulation: triangular numbers, finite sums, approximation, and area	Instructor model
Sep 2	1	Archimedes, geometric series, and the method of exhaustion	Colloquium begins
Sep 7	—	Labor Day — no class	—
Sep 9	1	Finite and infinite sums, sequences, convergence, computational experiments	Colloquium
Sep 14	1	Cavalieri, indivisibles, infinity, and the continuum	Colloquium
Sep 16	1	Partitioning, Riemann sums, sampling, pixels, and numerical approximation	Colloquium
Sep 21	2	The definite integral and accumulation functions	Colloquium
Sep 23	2	Barrow and the Fundamental Theorem of Calculus	Colloquium
Sep 28	2	Probability density, distributions, and expectation	Colloquium
Sep 30	2	Lorenz curves, inequality, and the Gini coefficient	Colloquium
Oct 5	2	Integration in economics: accumulation, supply, demand, and consumer surplus	Colloquium
Oct 7	2	Integration studio: data, visualization, modeling, and critique	Project I workshop
Oct 12	3	Motion, secants, finite differences, and instantaneous change	Colloquium
Oct 14	3	Leibniz, differential notation, and rules of differentiation	Colloquium
Oct 19	3	The derivative as limit: numerical, graphical, and analytic views	Colloquium
Oct 21	3	Products, quotients, composition, and the chain rule	Colloquium
Oct 26	3	Motion studies, photography, curves, and local linearity	Project I presentations
Oct 28	4	Critical points, maxima, minima, and objective functions	Project I presentations
Nov 2	4	Optimization, curvature, and objective landscapes	Colloquium
Nov 4	4	Newton's method: tangency as algorithm	Colloquium
Nov 9	4	Regression, error, loss, and the perceptron	Colloquium
Nov 11	4	Composition, the chain rule, neural networks, and back-propagation	Colloquium
Nov 16	5	From rates to differential equations: slope fields and numerical solutions	Colloquium
Nov 18	5	Euler's method, simulation, prediction, and model sensitivity	Colloquium
Nov 23	5	Growth, epidemics, populations, and nonlinear models	Colloquium
Nov 25	—	Thanksgiving — no class	—
Nov 30	6	Equilibrium, stability, feedback, cybernetics, and machine intelligence	Colloquium
Dec 2	7	From ordinary to partial differential equations: oscillation, waves, and the vibrating string	Colloquium
Dec 7	7	Surfaces, contours, partial derivatives, gradients, and fields	Colloquium
Dec 9	8	Vectors, embeddings, images, and machine representation	Final project workshop
Dec 14	8	Diffusion, fields, generative models, and representation	Final Project Symposium

Working Practices

Mathematics in this course will be developed through several different forms of work: calculation by hand, drawing and visualization, computational experimentation, mathematical argument, technical writing, historical reading, and discussion.

Students will use computational tools to investigate patterns, test conjectures, visualize mathematical objects, simulate models, and work with data. Depending on the problem, appropriate tools may include Jupyter and Python, spreadsheets, Desmos, GeoGebra, computer algebra systems, or other mathematical software. Students are expected to document computational experiments sufficiently for another person to understand and reproduce them.

Students will also learn to use $\text{\LaTeX}$ for mathematical and technical writing. Templates and examples will be provided, and expectations will develop over the course of the semester.

Generative Artificial Intelligence

Generative AI is part of the subject matter of this course, but it is not initially permitted as a tool for completing coursework.

During the semester we will develop the mathematical ideas needed to understand neural networks, loss functions, gradient-based learning, backpropagation, representation, and related computational systems. We will also examine these technologies historically and critically.

Later in the semester, the class will reconsider when and how generative AI systems may appropriately be used in coursework. Any change in this policy will be announced explicitly. Until then, generative AI systems should not be used to generate mathematical solutions, explanations, code, reading responses, or written work submitted for the course.

The governing principle is that we will work toward understanding these systems before relying on them as mathematical tools.

Assessment

Assessment in this course emphasizes sustained mathematical work, experimentation, communication, revision, and synthesis.

Course Work	Weight
Problem Sets & Computational Work	30%
Calculus Colloquium	20%
Project I: Modeling, Measurement & Representation	20%
Final Project & Symposium	25%
Participation, Discussion & Peer Feedback	5%

Problem sets will combine mathematical problems, computational experiments, visual reasoning, and occasional short responses to readings. Correct answers matter, but so do explanations, representations, documentation, and the ability to distinguish computational evidence from mathematical argument.

Instructor

Instructor: Jacob Koehler
Email: TBD
Office Hours: TBD

Reader

The readings in this course are not supplementary background to the mathematics. They are part of our investigation of what calculus is, how mathematical knowledge develops, how models and measurements acquire authority, and how ideas of change, optimization, representation, and intelligence have been transformed across different historical settings.

The reader moves among three kinds of texts:

- **Primary sources:** mathematical, scientific, and technical texts written by historical actors as ideas were being developed;
- **History and sociology of science:** scholarship examining mathematics, models, objectivity, quantification, and scientific practice historically and socially;
- **Theory and cultural studies:** texts that help us question space, classification, representation, images, technology, and artificial intelligence.

Most reading assignments are deliberately short. Read slowly and bring the text into contact with the mathematics. For each encounter, use the three questions below:

- READ** What is the author actually claiming, constructing, or trying to accomplish?
- CONNECT** What happens when we place the text beside the mathematics, computation, model, or image we are studying?
- QUESTION** Where should we challenge, test, complicate, or refuse the author's account?

Encounter 1

August 26

GENESIS: WHAT IS CALCULUS?

Otto Toeplitz, *The Calculus: A Genetic Approach*.

Preface to the English Edition and the opening section of Part I, "The Nature of the Infinite Process." (toeplitz1963)

Type: *History of mathematics / pedagogy*

- READ** What does Toeplitz mean by a "genetic approach" to mathematics? What problem with the usual presentation of calculus is he trying to address?
- CONNECT** Choose one mathematical idea from our opening problems. How did that idea emerge from a problem, construction, or experiment rather than from a definition or formula?
- QUESTION** Identify one claim Toeplitz makes about how mathematics should be learned or presented. What do you find convincing, questionable, or unresolved about that claim?

Encounter 2

September 2

ARCHIMEDES: MEASUREMENT, EXHAUSTION, AND MATTER

Archimedes, *Quadrature of the Parabola*. Short primary-source selection from the geometric argument.

Type: Primary mathematical source

Michel Serres, *The Birth of Physics*. “The Work of Archimedes,” working selection pp. 32–36, with a short selection from “Archimedes on Thinking Deviation.”

Type: Philosophy / history of science

READ How does Archimedes reason about an area without possessing our modern definition of the definite integral? What does Serres think is at stake in Archimedean mathematics?

CONNECT Identify the finite approximations in Archimedes’ argument. How are they like, and unlike, the computational approximations we construct in Jupyter?

QUESTION Does describing Archimedes as someone who “almost discovered calculus” help us understand his work, or erase the mathematical world in which he worked?

Encounter 3

September 14

INDIVISIBLES, THE CONTINUUM, AND NOMAD SCIENCE

Bonaventura Cavalieri, “Principle of Cavalieri” and “Integration,” in *struik1986*, selections beginning pp. 209 and 214.

Type: Primary mathematical source

Gilles Deleuze and Félix Guattari, “1227: Treatise on Nomadology—The War Machine,” in *A Thousand Plateaus*.

Selected passages on nomad science and royal or State science. (*deleuzeguattari1987*)

Type: Philosophy / cultural theory

READ How does Cavalieri reason with indivisibles? What distinction do Deleuze and Guattari draw between nomad and royal forms of knowledge?

CONNECT Does Cavalieri’s treatment of continuous magnitudes fit Deleuze and Guattari’s description of a mathematical practice concerned with becoming, variation, and local operations rather than fixed forms?

QUESTION Does “nomad science” actually illuminate Cavalieri’s mathematics, or are we imposing a twentieth-century philosophical category on a seventeenth-century practice?

Encounter 4

September 16

SMOOTH AND STRIATED SPACE

Deleuze and Guattari, *A Thousand Plateaus*. Short selected passage on smooth and striated space.

Type: Philosophy / cultural theory

READ How do smooth and striated spaces differ in the authors’ account?

CONNECT A Riemann sum imposes a coordinate system, partitions an interval, samples a function, assigns rectangles, and aggregates measurements. In what sense does this “striae” a continuous object?

QUESTION What becomes mathematically possible because of the partition? What information or experience of the continuous object might the partition suppress?

Encounter 5

September 23

BEFORE THE FUNDAMENTAL THEOREM WAS FUNDAMENTAL

Isaac Barrow, “The Fundamental Theorem of the Calculus,” in *struik*1986, selection beginning p. 253.*Type: Primary mathematical source*

READ What relationship does Barrow establish between problems involving tangents and problems involving areas?

CONNECT Translate Barrow’s geometric reasoning into our modern relationship between accumulation and differentiation:

$$F(x) = \int_a^x f(t) dt \quad \implies \quad F'(x) = f(x).$$

What does our notation make easier to see?

QUESTION What gets lost when we call Barrow’s argument “the Fundamental Theorem of Calculus” before the modern objects called derivatives and integrals have been stabilized?

Encounter 6

September 30

QUANTIFICATION: WHEN INEQUALITY BECOMES A NUMBER

Theodore M. Porter, *Trust in Numbers*. Working selection from “Cultures of Objectivity” and “How Social Numbers Are Made Valid.” (Porter 1995)

Type: History / sociology of quantification

READ Why, for Porter, can quantitative procedures acquire authority precisely by appearing to reduce personal judgment?

CONNECT Construct a Lorenz curve and calculate a Gini coefficient. Identify at least three decisions required before “inequality” can be represented as an area and then as a single number.

QUESTION Which decisions in the Gini calculation are mathematical and which are social? Is that distinction as clear as it first appears?

Encounter 7

October 14

THE DIFFERENTIAL CALCULUS APPEARS

Gottfried Wilhelm Leibniz, “The First Publication of His Differential Calculus,” in *struik*1986, selection beginning p. 271.

Type: Primary mathematical source

READ What mathematical objects and operations does Leibniz introduce? How does his notation participate in the argument?

CONNECT Translate a portion of Leibniz’s argument into contemporary notation. Identify where we would now speak about a derivative, differential, product rule, quotient rule, or tangent.

QUESTION Does translating Leibniz into modern notation help us understand him, or does it make his mathematics look more like ours than it really was?

Encounter 8**October 26****SEEING CHANGE: IMAGES AND OBJECTIVITY****Lorraine Daston and Peter Galison, *Objectivity*.**

Selected passages on mechanical objectivity and scientific images. (Daston and Galison 2007)

*Type: History of science / visual culture***Visual archive: selected motion studies by Eadweard Muybridge and Étienne-Jules Marey.***Type: Primary visual sources*

- READ** What does “mechanical objectivity” promise? What practices of seeing does it replace or reorganize?
- CONNECT** Compare a mathematical graph of motion, a sequence of photographs, a table of measurements, and a numerical derivative. What form of change does each make visible?
- QUESTION** Is an image evidence, measurement, interpretation, or model? Can it be all four?

Encounter 9**November 4****SLOPE, EXTREMA, AND OPTIMIZATION****Michel Serres, *The Birth of Physics*.**

Selected passage from “Slope and Extrema,” beginning around p. 52.

Type: Philosophy / history of science

- READ** How does Serres connect slope, deviation, equilibrium, and extrema to a physical picture of the world?
- CONNECT** If a function is a landscape, its derivative tells us which way the landscape slopes. An extremum is a place where that local direction vanishes.
What changes when we turn this geometric fact into an algorithm: repeatedly moving in the direction indicated by the derivative?
- QUESTION** Is optimization simply a mathematical technique, or does it encourage a particular way of imagining a problem—as a landscape with states, directions, extrema, and preferred paths?

Encounter 8**November 9****WHAT DOES IT MEAN FOR A MACHINE TO LEARN?****Frank Rosenblatt, “The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain.”** Selected opening and model passages from the 1958 paper.*Type: Primary source / history of artificial intelligence*

- READ** What does Rosenblatt claim the perceptron can model, and what does he mean by learning or self-organization?
- CONNECT** In our perceptron model, identify the inputs, weights, weighted sum, decision rule, error, and update. Exactly what changes when the machine “learns”?

QUESTION Formulate the strongest mathematical argument you can for calling parameter adjustment “learning,” and the strongest conceptual argument against doing so.

Encounter 9

November 11

THE CHAIN RULE BECOMES A LEARNING PROCEDURE

David E. Rumelhart, Geoffrey E. Hinton, and Ronald J. Williams, “Learning Representations by Back-Propagating Errors.” *Nature* 323 (1986), pp. 533–536.

Type: Primary technical source / artificial intelligence

READ What problem does backpropagation solve, and what do the authors claim hidden units learn to represent?

CONNECT Annotate the paper using the mathematics of the course: Where is the function? Where is composition? Where is the loss? Where is the derivative? Where is the chain rule? Where is optimization?

QUESTION What has to happen conceptually before a sequence of derivative calculations can be described as “learning representations”?

Encounter 10

November 16

FLOWS AND PATHS

Michel Serres, *The Birth of Physics*. Working selection from “Flows and Paths,” beginning around p. 70.

Type: Philosophy / history of science

READ What distinction does Serres make between describing an object and describing a flow, path, or process?

CONNECT A differential equation gives a rule for change rather than directly giving a trajectory. How does this alter what it means to know a mathematical object?

QUESTION Does the language of flow illuminate differential equations, or become merely metaphorical once we formalize the mathematics?

Encounter 11

November 18

CAN A MODEL BE VERIFIED?

Naomi Oreskes, Kristin Shrader-Frechette, and Kenneth Belitz, “Verification, Validation, and Confirmation of Numerical Models in the Earth Sciences.” *Science* 263 (1994), pp. 641–646.

Type: Philosophy / history of modeling

READ Why do the authors distinguish confirmation from verification or validation for models of open natural systems?

CONNECT Return to an Euler-method simulation. Which parts come from mathematics, which from empirical assumptions, which from numerical choices, and which from initial conditions?

QUESTION If a model produces accurate predictions, what has actually been established? What has not?

Encounter 12

November 30

FEEDBACK, CONTROL, AND INTELLIGENCE

Norbert Wiener, *The Human Use of Human Beings*. Short selection on cybernetics, feedback, communication, and purposive behavior. (Wiener 1950)

Type: Primary source / cybernetics

Andrew Pickering, *The Cybernetic Brain*. Short historical selection on cybernetic practice.

Type: History / sociology of science

- READ** How does feedback permit Wiener and other cyberneticians to describe purposive behavior without beginning from an internal mind or intention?
- CONNECT** Identify feedback mathematically in one of our dynamical systems. How do equilibrium and stability change when the output of a system becomes part of its future input?
- QUESTION** What is gained—and what may be lost—when intelligence, behavior, or purpose is redescribed in terms of information and feedback?

Encounter 13

December 2

WHEN AN EQUATION BECOMES A FIELD

Jean le Rond d'Alembert, Leonhard Euler, and Daniel Bernoulli, "The Vibrating String and Its Partial Differential Equation," in struik1986, selection beginning p. 351.

Type: Primary mathematical / physical source

- READ** What is being represented mathematically when the vibrating string is described not by a single changing quantity but by a function of both position and time?
- CONNECT** Compare

$$y = y(t)$$

with

$$u = u(x, t).$$

Why does the second description require partial derivatives? What does each independent variable represent?

- QUESTION** Is the partial differential equation a description of the string, a model of the string, or a new mathematical object with a life independent of the physical system that produced it?

Encounter 14

December 7

SURFACES, DIMENSIONS, AND CLASSIFICATION

Gaspard Monge, "The Two Curvatures of a Curved Surface," in struik1986, short selection beginning p. 413.

Type: Primary mathematical source

Geoffrey C. Bowker and Susan Leigh Star, *Sorting Things Out: Classification and Its Consequences*.

Selected introduction. (Bowker and Star 1999)

Type: Science and technology studies

READ How does moving from a curve to a surface change the mathematical problem? How do Bowker and Star describe the work performed by systems of classification?

CONNECT For a curve

$$y = f(x),$$

local change can be described by a single derivative. For a surface

$$z = f(x, y),$$

change depends upon direction.

How does moving from curves to surfaces change the meaning of slope, tangent, and curvature? How does this prepare us to think about high-dimensional spaces in which objects are represented by many coordinates?

QUESTION An embedding represents an object as a point in a high-dimensional space. What decisions must occur before similarity, difference, and classification can appear as geometric relationships?

Encounter 17

December 9

REPRESENTATION, ATTENTION, AND MACHINE VISION

Ashish Vaswani et al., “Attention Is All You Need.”

Selected passages and diagrams introducing embeddings, self-attention, and the Transformer architecture.

Type: Primary technical source / artificial intelligence

Trevor Paglen, “Invisible Images (Your Pictures Are Looking at You).”

Type: Art / visual culture / critical technology studies

READ What mathematical objects enter a Transformer? What operation does attention perform on a collection of representations?

What changes, for Paglen, when images and representations circulate primarily among machines rather than between images and human viewers?

CONNECT Trace two related transformations:

token $\longrightarrow$ vector $\longrightarrow$ embedding $\longrightarrow$ attention $\longrightarrow$ representation

and

image → pixels → numbers → features → embedding.

Where do the mathematical ideas we have studied this semester appear in these processes?

QUESTION When we say that a model “attends” to something, what mathematical operation are we describing?

What is gained by this language, and what might it encourage us to imagine about the machine that is not contained in the mathematics?

Encounter 18

December 14

GENERATIVE AI: MATHEMATICS, INFRASTRUCTURE, AND CULTURE

Kate Crawford, *Atlas of AI*. Selected concluding or synthetic passage. (Crawford 2021)

Type: Critical AI / sociology of technology

Contemporary primary technical artifact. A short excerpt, diagram, or model description from work on diffusion or generative models, to be selected.

Type: Primary technical source

READ What material, institutional, political, and labor systems does Crawford ask us to see when we use the term “artificial intelligence”?

CONNECT Our mathematical story may describe a generative system through probability, gradients, fields, differential equations, or optimization. What does that description explain extremely well? What does it leave outside the frame?

QUESTION What would an adequate account of generative AI have to contain in addition to its mathematics?

Calculus Colloquium: Extended Reader

The shared reader is intentionally small. The Calculus Colloquium expands it. Students will select additional primary sources, historical scholarship, theoretical texts, mathematical problems, visual artifacts, datasets, or technical papers related to the current module.

Possible directions include Archimedes, Lucretius, Cavalieri, Newton, Leibniz, Euler, Fourier, Marey, Muybridge, Turing, Shannon, McCulloch and Pitts, Hebb, Rosenblatt, Wiener, Minsky and Papert, Rumelhart–Hinton–Williams, Hopfield, convolutional networks, word embeddings, attention, transformers, GANs, diffusion models, cybernetic art, Project Cybersyn, ImageNet, algorithmic classification, scientific visualization, and histories of computation.

A colloquium contribution should not simply summarize its source. It should make an argument about the relationship between the source and the mathematics of the course.